

Yi Wang

AI Researcher · Representation, Predictive State & Intelligent Systems

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📄 arXiv: 2608.18025

🎵 Kinssion / Music

🏢 Department of Electronic Engineering, Tsinghua University



M.S. Student · Advisor: Prof. Linglong Dai



RESEARCH PROFILE

M.S. student in Electronic Engineering at Tsinghua University studying **representation, predictive state, and revisable intelligence** in AI systems. My research asks how static representations should divide predictive work with contextual state, what information a system must preserve for future prediction, and how that state can remain recursively updatable as queries, evidence, and mechanisms change. In parallel, I study physics-grounded wireless systems through robust optimization, mechanism discovery, and structured beam design.

Representation → Predictive State → Recursive Memory → Revisable Intelligence

Research style: Problem discovery · Theory & modeling · Model/algorithm design · Controlled experiments · End-to-end implementation



AI RESEARCH

How Far Should Tokenization Go? Predictive Effectiveness and Relational Losslessness

2026

Sole Author · Under Review, ICLR 2027

What reusable structure should be explicit in a representation, and what must remain for a model to infer from context?

- **Representation-State Handoff:** Reframe tokenization from sequence compaction to a division of predictive labor between static representation and contextual state.
- **Effectiveness-Losslessness Framework:** Introduce Predictive Effectiveness and Relational Losslessness, together with the Fact-Token and Token-State boundaries, to determine how far static representation should proceed.
- **Controlled evidence:** Built the full data, model, and evaluation pipeline for a **9-arm × 3-seed** study. Musical-time coordinates improve predictive codelength by **0.848 bits/fact (11.92%)**; reversible BPE reduces serialization from **3.563 to 1.538 targets/fact (56.8%)** yet increases predictive codelength by **0.811 bits/fact**.

From Source Reconstruction to Predictive State Preservation: An Information-Theoretic Framework for AI-Native Communication

2026

First Author · Manuscript

What is the minimal information an intelligent system must preserve for future prediction?

- **Minimal predictive state:** Identify $S_X = P(F \mid X, C)$ as the minimal prediction-preserving information object: distinctions in the raw observation that do not alter the conditional future law need not be preserved.
- **From static state to memory:** Show that fixed-horizon predictive sufficiency does not guarantee recursive closure; the full-future predictive state provides a minimal recursively sufficient state with Markov evolution.

From Predictive State Preservation to Behavioral Fidelity: Query-Update Information Objects for Revisable Communication

2026

First Author · Manuscript

What must a system preserve when its questions, evidence, actions, and mechanisms may change?

- **Behavioral and update-stable state:** Extend fidelity from a fixed prediction target to query-update behavior and construct an update-stable quotient; under partial observation, the recursive information object becomes a quotient belief.
- **Plasticity beyond state:** Separate four levels—**Behavior** → **State** → **Relational Realization** → **Code/Patch Plasticity**—to distinguish what behavior must survive from how internal representations are realized and revised.

Behavior defines fidelity; update closure defines state; structure realizes state; code determines plasticity.

Four first-author works developed within four months

Robustness → Physics-Guided AI → Mechanism → New Beam Paradigm

- **Robustness:** Proposed Multi-Airy trajectory diversity, subarray decomposition, and coherent combining, reducing rate loss under a 40-mm geometry error from **97.2% to 33.3%**.
- **Physics-Guided AI and mechanism:** Converted trajectory–edge coupling into a one-shot neural predictor with about **112× fewer parameters** while retaining **99.7%** of reference performance. Reinterpreted Airy recovery as hierarchical compensation of blockage-induced phase mismatch: the curved trajectory is a propagated consequence, while the cubic term closes the residual to **99.77%** of the phase-only bound with an $\mathcal{O}(N)$ one-shot realization.
- **New beam paradigm:** Developed Channel-Matched Structured Beams / Edge-Matched Beams as low-dimensional reusable beam families that approximate channel-optimal solutions; a four-beam realization gains **3.06 dB** and **0.64 bit/s/Hz**.

 CREATION & ZERO-TO-ONE

- **Independent Musician (Kinssion):** 50+ original releases and 770k+ plays; built the full creation-to-production pipeline.
- **Founder, Tsinghua Student Original Music Club:** Founded Tsinghua’s first student original-music community, now with 200+ members.
- **Embedded Music Box:** Built an interactive composition hardware prototype; received a department hardware-design award.

 EDUCATION

Tsinghua University

2024–Present

M.S. in Electronic Engineering · Advisor: Prof. Linglong Dai

Tsinghua University

2020–2024

B.Eng. in Electronic Information Science and Technology
Music Engineering / Music Theory & Composition Program, 2021–2023

 TECHNICAL SKILLS

AI / ML	PyTorch · Transformers · Autoregressive Modeling · Representation Learning · Generative Modeling
Theory	Information Theory · Predictive Sufficiency · Rate–Distortion · Optimization · Probabilistic Modeling
Research Engineering	Python · MATLAB · C/C++ · Dataset & Training Pipelines · Experiment Automation · Numerical Simulation

 SELECTED RESEARCH / PUBLICATIONS

First author on all listed works; sole author of the music study.

AI & Information Theory

1. **How Far Should Tokenization Go? Predictive Effectiveness and Relational Losslessness.** *Under Review, ICLR 2027.*
2. **From Source Reconstruction to Predictive State Preservation: An Information-Theoretic Framework for AI-Native Communication.** *Manuscript, 2026.*
3. **From Predictive State Preservation to Behavioral Fidelity: Query–Update Information Objects for Revisable Communication.** *Manuscript, 2026.*

Wireless & Physical Systems

4. **Blockage-Robust Beamforming for Near-Field Communications: From Single-Airy to Multi-Airy.** *Submitted, IEEE TWC, 2026.*
5. **Physics-Guided Neural Airy Beamforming for Near-Field Blockage Mitigation.** *Submitted, IEEE JSAC, 2026.*
6. **Why Is Cubic-Phase Airy Beamforming Sufficient for Blockage Recovery?** *Manuscript, 2026.*
7. **Beyond Airy Beams: Channel-Matched Structured Beams for Near-Field Blockage Mitigation.** *Manuscript, 2026.*